NeurIPS 2026 REINFORCEMENT LEARNING · GENERATIVE POLICIES

Generative Actor-Critic with Soft Bridge Policies

One step outside, a **soft bridge** inside: a principled MaxEnt actor that beats many-step diffusion in a single pass.

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NEURAL INFORMATION PROCESSING SYSTEMS

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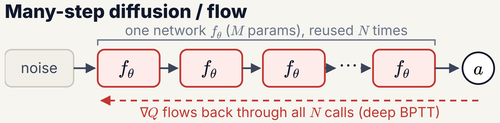
Code (JAX)
Scan for the repo
github.com/skypitcher/soft_gac_jax
arXiv:2605.08733

1 Two problems

Why generative? They capture **multimodal** actions, e.g. going left *or* right around an obstacle.

PROBLEM 1 **Cost vs. performance**

Many-step diffusion / flow — NFE = N



✓ strong returns
✗ N× inference cost
✗ deep, unstable BPTT

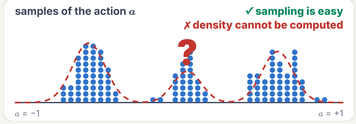
One-step generator (black box) — NFE = 1



✓ one network call
✗ returns usually drop

PROBLEM 2 **Intractable entropy**

For **any implicit generator** (one step or many): easy to sample, but no tractable density or entropy.



$\pi_\theta(a \mid s) = \int p(z)\,\delta(a - f_\theta(z,s))\,dz \;\Rightarrow\; \mathcal{H}(\pi_\theta) = ?$

Approximate the entropy INEXACT
GMM fit · Monte Carlo · lower bound

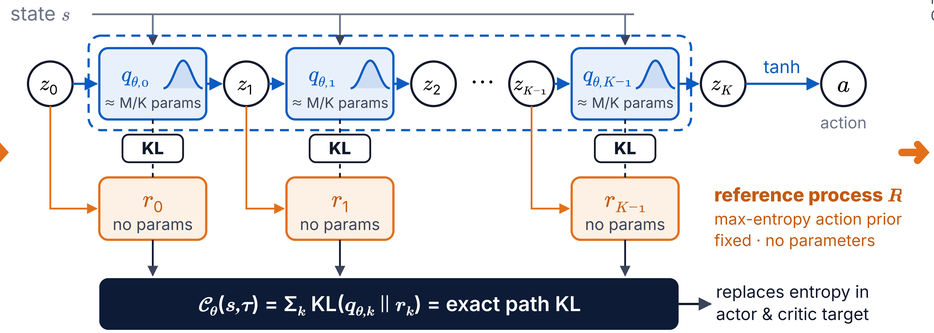
Replace it with a surrogate INDIRECT
kinetic energy · Wasserstein distance

Neither guarantees endpoint-equivalence.

Can one policy be truly soft, fast, and better?

2 Open the box: a soft bridge

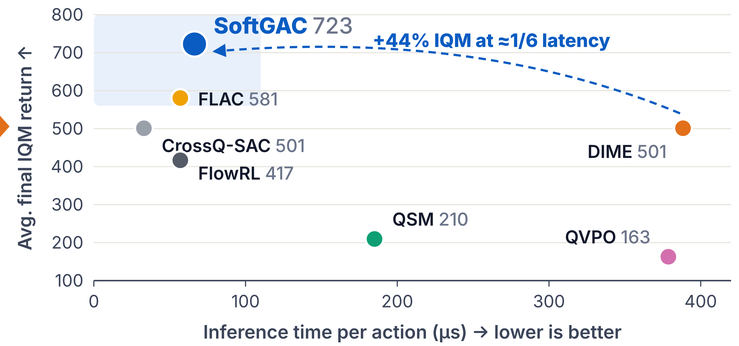
About the **same M params**, still **one call** (the dashed box is f_θ). Inside, the budget is split into K small Gaussian layers (**≈ M/K params** each), each matched by one step of a fixed max-entropy reference process (no params). Their KLs are closed form and add up to the **exact** soft regularizer, computed in the same pass.



3 Faster, yet stronger

+44%
avg. IQM over the strongest many-step actor (DIME), with 0.43M vs. 0.44M actor params

≈1/6
of DIME's per-action inference latency



Mean final IQM on the 8 hard tasks vs. per-action latency (mean of 3 domains, M3 Pro CPU), from Figs. 3 and 9 of the paper. Also +24% over the best baseline overall (FLAC).

4 Why it is principled: MaxEnt in path space

SAC regularizes the action law: $\mathbb{E}_\pi[Q] - \alpha D_{\mathrm{KL}}(\pi \,\|\, u)$. For any implicit generator, $\pi_\theta(a|s)$ is intractable, so prior methods approximate the entropy (inexact) or use surrogates (indirect). We regularize the **whole path** $\tau = (z_0, \ldots, z_K)$ instead:

$$\mathcal{J}_{\mathrm{path}} = \mathbb{E}_{\tau\sim P_\theta}\big[Q(s, \tanh z_K)\big] - \alpha\, D_{\mathrm{KL}}\big(P_\theta(\cdot \mid s) \,\|\, R\big)$$

PROP. 1 · PATH KL CONTAINS SAC'S REGULARIZER

$$D_{\mathrm{KL}}(P_\theta \| R) = \underbrace{D_{\mathrm{KL}}(\pi_\theta \| u_R)}_{\text{endpoint KL (SAC)}} + \underbrace{\mathbb{E}_{a\sim\pi_\theta}\big[D_{\mathrm{KL}}(P_\theta(\cdot|a) \| R(\cdot|a))\big]}_{\text{how } a \text{ is generated}}$$

IDEAL THM. 1 · UNCONSTRAINED BASE

Over all path laws $P \ll R$, *including* the law of z_0:

$$\frac{dP_Q^\star}{dR}(\tau \mid s) \propto e^{Q(s,\tanh z_K)/\alpha} \implies \pi_Q^\star(a \mid s) \propto u_R(a)\, e^{Q(s,a)/\alpha}$$

ENDPOINT-EQUIVALENCE With uniform u_R, $\pi_Q^\star$ is **exactly SAC's Boltzmann policy**. The path term only shapes *how* each action is reached.

PRACTICAL THM. 2 & COR. 1 · FIXED BASE

The ideal optimum also reweights z_0 by $Z_Q(s, z_0)$, a state-dependent base sampler. A real actor fixes p_0 (ours: q_{ref}), so only the transitions are tilted:

$$P^\star_{Q,p_0}(d\tau \mid s) = p_0(dz_0)\, \frac{e^{Q(s,\tanh z_K)/\alpha}}{Z_Q(s,z_0)}\, R(d\tau_{1:K} \mid z_0)$$

The base adds only a constant, and its gap $\Delta(s) \geq 0$ bounds the endpoint error: $D_{\mathrm{KL}}(\pi^\star_{Q,p_0} \| \pi^\star_Q) \leq \Delta(s)/\alpha$.

5 How it is built and trained

Each layer, and its reference step

$$q_{\theta,k}(z_{k+1} \mid z_k, s) = \mathcal{N}\big(z_k + h\, u_{\theta,k}(s, z_k),\ 2h\, \mathrm{diag}\, \sigma^2_{\theta,k}\big)$$

$$z_0 \sim q_{\mathrm{ref}} \propto \textstyle\prod_i \mathrm{sech}^2 z_i, \qquad r_k = \mathcal{N}\big(z_k - 2h \tanh z_k,\ 2hI\big)$$

$h = 1/K$. Each layer is its own small MLP (≈ M/K params; about M in total), not a shared sampler. $\tanh$ maps q_{ref} to the uniform law, so R is a max-entropy action prior.

LEMMA 1 · EXACT CONTROL ENERGY

$$\mathcal{C}_\theta(s,\tau) = \sum_{k=0}^{K-1} \frac{1}{2} \sum_{i=1}^{d_a} \left[\sigma^2_{\theta,k,i} + \frac{(\mu_{\theta,k,i} - \mu_{R,k,i})^2}{2h} - 1 - \log \sigma^2_{\theta,k,i}\right]$$

with $\mathbb{E}_{\tau\sim P_\theta}[\mathcal{C}_\theta] = D_{\mathrm{KL}}(P_\theta \| R)$: closed form, no estimation.

Drop-in soft actor-critic

Endpoint entropy is replaced by control energy, used **directly** in both updates:

$$y = r + \gamma(1-d)\Big[Q^{\min}_{\bar\phi}\big(s', \tanh z'_K\big) - \alpha\, \mathcal{C}_{\bar\theta}(s', \tau')\Big]$$

$$\mathcal{L}_{\mathrm{actor}}(\theta) = \mathbb{E}_{s,\ \tau\sim P_\theta}\Big[\alpha\, \mathcal{C}_\theta(s,\tau) - Q^{\min}_\phi\big(s, \tanh z_K\big)\Big]$$

PROP. 2 · A REVERSE-KL PROJECTION

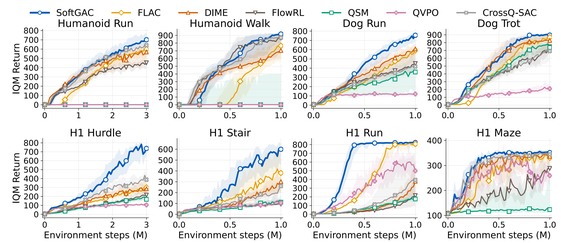
$$\mathcal{L}_{\mathrm{actor}}(\theta \mid s) = \alpha\, D_{\mathrm{KL}}\big(P_\theta \,\|\, P^\star_Q\big) + \mathrm{const}$$

α tracks a control budget $\mathcal{C}_{\mathrm{target}} = \rho_{\mathrm{ctrl}} K d_a$. Deployment follows the mean bridge path.

6 Results: faster does not have to mean weaker

DMC (Humanoid, Dog) + HumanoidBench (H1) · 12 tasks · 8 seeds · IQM, 95% CI · all actors re-implemented in one codebase with comparable parameter budgets.

Learning curves on the 8 hard tasks



Largest gains on high-dimensional, long-horizon locomotion (Humanoid Run, Dog Run, H1).

vs. 20-step diffusion actors · H1 Hurdle (3M)

Actor	NFE	Params	IQM	Inference	Train
SoftGAC	**1**	**0.34M**	**737**	**63 μs**	**6.0 h**
DACER	20	0.32M	377	400 μs	7.6 h
SDAC	20	0.32M	60	410 μs	9.3 h
MaxEntDP	20	0.40M	18	509 μs	24.9 h

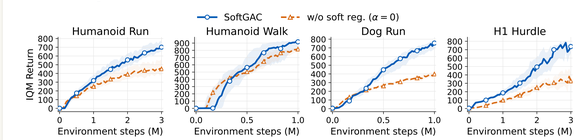
vs. a one-step flow with exact density

SoftGAC	724
Normalizing flow	478
DACER (20 NFE)	432

Average IQM over Dog Run, H1 Hurdle, H1 Run, H1 Stair and Humanoid Run. Flows buy exact density with invertible layers; the bridge needs neither invertibility nor Jacobians.

7 Why it works

The soft path term matters



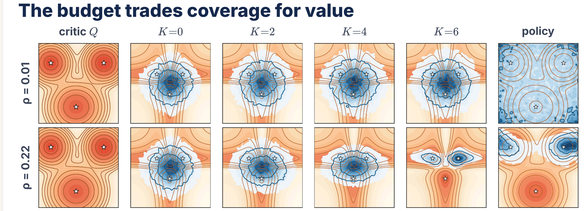
Same actor and critic with $\alpha = 0$: removing the control energy consistently hurts.

Analytical KL » estimated entropy

Path KL (ours)	729
GMM entropy	349

Same actor with DACER's GMM endpoint-entropy estimator. Avg. IQM, 4 tasks.

The budget trades coverage for value



2D toy, 3 value modes. Real policies are multimodal: **53.8%** of H1 Hurdle states (CrossQ-SAC: 0%). Also robust to base, depth and critic (XQC).

Takeaways

① **Soft** Analytical path-KL, endpoint-equivalent to SAC. No estimates or surrogates.

② **Fast** One call per action, about the same params: ≈1/6 the latency.

③ **Better** +44% return over the strongest many-step policy.

Limitations: extra design choices; state-based simulation only.